

Personal Loan Ledger Mobile Application

Software Requirements Specification (SRS)

Shared Loan Tracking, Reminders, Repayment Confirmation, Dashboard Analytics, and Bank Profile Sharing

Document	Software Requirements Specification (SRS)
Version	Version 1.0 - Product Foundation
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Purpose	Defines the complete functional and non-functional requirements for the initial production-ready mobile application.

Product scope note: The application tracks peer-to-peer loans and repayment acknowledgements. It does not custody funds, execute bank transfers, provide credit scoring, or automatically enforce legal debt collection in the initial release.

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1. Purpose and Scope

This SRS defines what the mobile application must do, who it serves, the business rules governing shared personal loans, and the quality attributes required for a secure and reliable release.

1.1 In-Scope Product

- Cross-platform React Native mobile application for iOS and Android.
- Account creation, identity profile, friend connections, shared loan records, repayment confirmations, dashboards, reminders, and bank-profile sharing.
- A backend service and database that act as the authoritative source of truth for both participants in a loan.
- Push notifications for requests, approvals, due dates, overdue loans, repayment submissions, and confirmations.
- No movement or custody of money by the platform in the MVP.

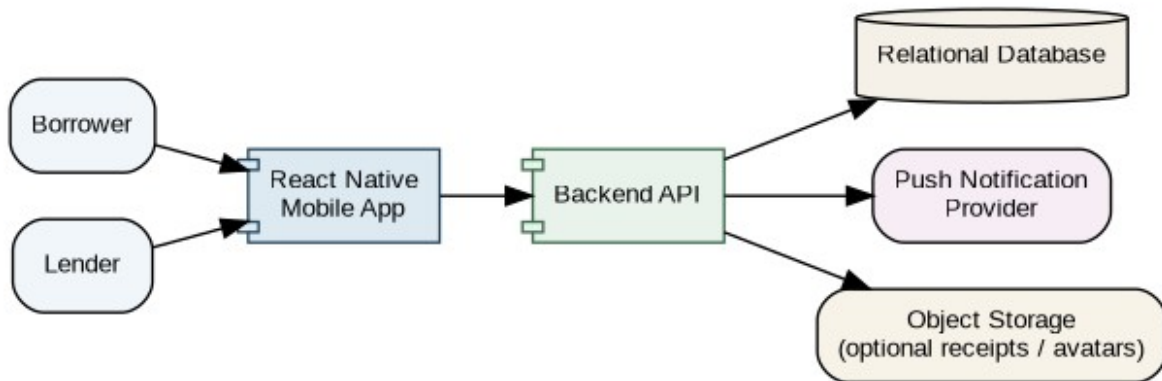


Figure 1. System context - users interact with a shared backend through the React Native application.

1.2 Definitions

Term	Definition
Borrower	The user who receives or requests money and is responsible for repayment.
Lender	The user who provides money and confirms that repayment has been received.
Loan request	A borrower-originated request to create a shared debt record with a selected lender.
Loan terms	Principal, currency, interest arrangement, due date, notes, and acknowledgements accepted by both parties.
Bank profile	A user-managed payment destination such as bank or mobile-wallet account details shared only with authorized counterparties.
Repayment submission	A borrower claim that all or part of a loan has been repaid, awaiting lender confirmation.

2. Product Vision and Business Objectives

The product exists to replace informal paper notes, private chat messages, and memory-based tracking with a shared and auditable record that both parties can see.

- **Primary objective** - Make personal lending records clear, shared, timestamped, and easy to follow.
- **Trust objective** - Require explicit acceptance for loan terms and repayment completion so neither party can silently rewrite history.
- **Visibility objective** - Give each user a dashboard that explains money owed to them, money they owe, due dates, overdue exposure, and pending actions.
- **Reminder objective** - Reduce missed repayment dates with configurable push notifications.
- **Payment coordination objective** - Allow users to securely share a preferred bank or wallet profile for repayment instructions without turning the application into a payment processor.

2.1 Success Measures

Metric	Initial Target / Interpretation
Loan confirmation completeness	100% of active loans have explicit lender acceptance and stored terms.
Repayment confirmation integrity	A loan cannot become completed until lender confirmation or an administrative resolution path is used.
Reminder delivery	Scheduled due-date reminders are queued and delivery outcome is logged.
Dashboard consistency	Dashboard totals reconcile with open loan balances by currency.
Security	No bank account identifier is exposed to a user who is not an authorized recipient.

3. Stakeholders and User Classes

Stakeholder	Needs / Responsibilities
Borrower	Request a loan, review terms, track what is owed, receive reminders, share repayment claim, view lender payment instructions.
Lender	Receive requests, set/confirm terms, track receivables, share preferred bank profile, confirm repayment.
Support/Admin	Investigate account problems, notification failures, disputes, abuse, and data integrity without silently changing financial history.
Product Owner	Define policy, scope, release rules, legal language, and analytics.
Engineering	Deliver secure client, APIs, background jobs, database integrity, and observability.
Security/Privacy	Define data retention, encryption, access control, incident handling, and privacy obligations.

3.1 User Roles

- Standard authenticated user - may act as borrower in one loan and lender in another.
- Support agent - restricted operational access; no ability to impersonate financial acknowledgement without audit.
- Administrator - manages abuse, account status, feature flags, and exceptional data repair through audited workflows.

4. Assumptions, Constraints, and Dependencies

- Users must have internet access for synchronization and actions that change shared state.
- A valid push token is helpful but not mandatory; in-app notification history remains available.
- The initial release does not verify legal identity, creditworthiness, bank ownership, or whether money actually changed hands.
- Interest legality differs by jurisdiction. The product records user-agreed terms and must not present itself as legal or financial advice.
- Bank profiles are informational repayment destinations. The system does not initiate transfers in the MVP.
- Amounts are stored in the smallest practical decimal precision for each currency and never as binary floating-point values.
- Totals across different currencies must remain separated unless a future FX conversion feature is introduced.

5. User Journeys and Primary Workflows



Figure 2. Primary borrower-requested loan and repayment-confirmation workflow.

5.1 Happy Path

1. A user registers, verifies the account, and completes a basic profile.
2. The user connects with another user through a friend request or invite.
3. The borrower selects a friend and sends a loan request with an optional requested amount and note.
4. The lender reviews the request, sets or confirms principal, currency, flat interest rate, due date, and optional note, then accepts.
5. The loan becomes active and both parties see the exact agreed terms.
6. The system schedules due-soon and due-date notifications for both parties.
7. The borrower can view the lender's bank profile if the lender has shared one for the loan.
8. After payment outside the app, the borrower submits a repayment claim.
9. The lender confirms receipt; the loan becomes completed and remains available in history.

5.2 Alternate Flows

- Lender rejects a loan request.
- Borrower cancels a pending request before lender acceptance.
- Lender proposes changed terms; borrower must acknowledge material changes before activation if two-party term acknowledgement is enabled.
- Due date passes; status becomes overdue while remaining repayable.
- Repayment submission is rejected by lender with a reason; loan returns to active/overdue.
- Friend connection is removed after loan creation; existing loan remains visible to both parties until resolved.

6. Account Registration and Authentication Requirements

ID	Requirement
AUTH-001	The system shall allow account registration using email and/or phone number as configured by deployment.
AUTH-002	The system shall verify ownership of the selected login identifier before enabling loan actions.
AUTH-003	The system shall support secure sign-in, sign-out, token refresh, and session revocation.
AUTH-004	The system shall provide a password-reset or equivalent recovery flow.
AUTH-005	The system shall prevent disabled or deleted accounts from initiating new financial records.
AUTH-006	The system shall rate-limit authentication attempts and sensitive verification operations.
AUTH-007	The app shall require re-authentication for high-risk actions when the session is stale, including revealing full bank account identifiers or changing security settings.
AUTH-008	The backend shall authorize every resource request; client-side visibility alone shall never be treated as access control.

Security principle: Every loan, repayment, bank share, and friend relationship is server-authorized using the authenticated user ID; request-supplied user IDs are not trusted.

7. Profile Management Requirements

ID	Requirement
PRO-001	Users shall manage display name, username/handle if enabled, profile photo, default locale, timezone, and optional default currency.
PRO-002	The system shall support account-status states including active, suspended, deletion_pending, and deleted.
PRO-003	A user shall be able to configure reminder preferences without disabling mandatory in-app records of critical loan events.
PRO-004	Profile visibility shall be limited to the minimum needed for user discovery and connected-friend interactions.
PRO-005	Users shall be able to request account deletion; active unresolved loans require a defined retention/closure policy rather than destructive removal of shared records.
PRO-006	Profile changes that affect identity presentation shall not rewrite historical actor IDs in the loan event log.

8. Friend and Connection Management

ID	Requirement
FRI-001	A user shall search or invite another user using approved identifiers such as username, phone, email, or shareable invite link.
FRI-002	A friendship shall require recipient acceptance before normal loan requests can be sent, unless invite-based onboarding directly establishes consent.
FRI-003	Friendship states shall include pending, accepted, rejected, blocked, and removed.
FRI-004	Blocking shall prevent new requests and sensitive profile access.
FRI-005	Removing or blocking a friend shall not delete active or historical loans between the parties.
FRI-006	The app shall show per-friend financial summary by currency: amount they owe the user and amount the user owes them.
FRI-007	Duplicate accepted friendships between the same pair shall be prevented by a database constraint or canonical-pair strategy.

9. Loan Request and Acceptance Requirements

ID	Requirement
LOAN-001	A borrower shall be able to select an accepted friend and create a loan request.
LOAN-002	The request may include requested principal, requested currency, desired due date, and note; the lender remains responsible for the accepted terms.
LOAN-003	The lender shall be able to accept, reject, or return the request with changed proposed terms.
LOAN-004	Before acceptance, the lender shall specify principal, currency, interest configuration, due date, and optional description.
LOAN-005	The backend shall validate that borrower and lender are distinct users and have permission to interact.
LOAN-006	Acceptance shall create an immutable version of the accepted loan terms and a corresponding audit event.
LOAN-007	A loan shall not be considered active until the defined acceptance conditions have been satisfied.
LOAN-008	The system shall prevent unilateral edits to accepted principal, currency, interest, or original due date; modifications must be represented as amendments with history.
LOAN-009	A lender shall not confirm a loan request that has expired, been cancelled, or already been resolved.
LOAN-010	Each loan shall have a human-friendly reference code in addition to its internal UUID.

10. Loan Terms, Currency, Interest, and Due-Date Rules

10.1 Currency Rules

- Each loan has exactly one ISO 4217 currency code or an explicitly supported local currency code.
- Principal, interest, repayment, and balance calculations for a loan stay in that currency.
- Dashboard sums are grouped by currency.
- A default currency is a user convenience only; it never silently converts an existing loan.

10.2 Interest Model for MVP

- Interest may be zero.
- If percentage interest is used, the MVP interprets the value as a flat percentage applied once to principal for that loan, e.g., 5% on 10,000 ETB produces 500 ETB interest.
- The UI must label the basis clearly as “Flat loan interest %” to avoid confusing it with APR.
- The data model reserves fields for future APR, fixed-fee, compounding, and installment calculation methods.

ID	Requirement
TERM-001	Principal must be greater than zero and within configured maximum precision and amount limits.
TERM-002	Due date must be after the agreement time unless an administrative import flow is used.
TERM-003	Interest rate shall be validated against configured product limits and displayed with its calculation basis.
TERM-004	The total expected repayment shall be calculated server-side and reproduced client-side for preview.
TERM-005	All monetary calculations shall use decimal arithmetic and deterministic rounding rules.
TERM-006	The accepted terms snapshot shall store timezone-aware timestamps and the users' relevant display timezone separately where needed.

11. Loan Lifecycle and State Model

State	Meaning	Allowed Next States
REQUESTED	Borrower submitted; lender has not accepted.	REJECTED, CANCELLED, TERMS_PROPOSED, ACTIVE
TERMS_PROPOSED	Lender changed terms and borrower acknowledgement is required if enabled.	ACTIVE, CANCELLED, REJECTED
ACTIVE	Accepted and not fully confirmed repaid.	REPAYMENT_PENDING, OVERDUE, CANCELLED_BY_AGREEMENT
OVERDUE	Due date has passed with outstanding balance.	REPAYMENT_PENDING, COMPLETED, DISPUTED
REPAYMENT_PENDING	Borrower claims payment; lender confirmation pending.	COMPLETED, ACTIVE, OVERDUE, DISPUTED
COMPLETED	Lender confirmed required repayment.	Terminal except administrative annotation.
REJECTED	Lender rejected request.	Terminal.
CANCELLED	Pending request or mutually cancelled loan.	Terminal.
DISPUTED	Counterparties disagree or support intervention requested.	ACTIVE, OVERDUE, COMPLETED, CLOSED_DISPUTE

Invariant: Historical terminal events remain preserved even if the current display status changes through a support resolution. The event log is append-only.

12. Repayment Submission and Confirmation

ID	Requirement
PAY-001	The borrower shall be able to submit a repayment claim for an active or overdue loan.
PAY-002	The repayment submission shall capture amount, currency inherited from loan, submitted timestamp, optional note, and optional proof attachment if enabled.
PAY-003	For MVP full-repayment mode, the app shall default the claimed amount to the outstanding expected balance.
PAY-004	The lender shall receive an in-app and push notification for a repayment submission.
PAY-005	The lender shall explicitly confirm or reject a repayment submission.
PAY-006	Confirmation shall mark the repayment confirmed and close the loan when no balance remains.
PAY-007	Rejection shall require an optional/required reason according to policy and return the loan to active or overdue state.
PAY-008	The borrower shall not be allowed to confirm their own repayment claim as received.
PAY-009	The system shall prevent duplicate concurrent repayment claims that would exceed the outstanding balance.
PAY-010	The underlying schema shall support partial repayments even if the initial UI exposes full repayment only.

13. Preferred Bank Profile and Sharing Requirements

- A user may create multiple payment/bank profiles and designate one as preferred.
- A profile may represent a bank account, mobile wallet, or other supported repayment destination.
- Typical fields: bank/provider name, account holder name, account identifier, optional branch, optional routing/SWIFT/IBAN metadata, country, currency hints, and label.
- Sensitive account identifiers must be encrypted at rest and masked by default in UI.
- A bank profile is shared only through an explicit share record with a specific user and optionally a specific loan.
- The lender can attach/share the preferred profile when a loan becomes active or later from the loan details screen.
- The borrower receives the shared repayment instructions and may copy them after an explicit reveal action.
- The lender can revoke a share; revocation does not erase the audit record that sharing occurred.

ID	Requirement
BANK-001	Users shall create, edit, archive, and select a preferred bank/payment profile.
BANK-002	Only the profile owner and explicitly authorized recipients shall retrieve sensitive fields.
BANK-003	Account identifiers shall be encrypted using an application-managed or cloud KMS-backed key strategy.
BANK-004	Logs, analytics events, crash reports, and push-notification bodies shall never contain full account identifiers.
BANK-005	Sharing shall be revocable and auditable.
BANK-006	Bank-profile sharing shall not imply that the platform validated account ownership or initiated a transfer.

14. Dashboard and Financial Metrics

14.1 Dashboard Cards

Metric	Definition
Money others owe me	Sum of outstanding principal + applicable interest on receivable loans, grouped by currency.
Money I owe others	Sum of outstanding payable balances, grouped by currency.
Net position	Receivables minus payables within the same currency only.
Active receivable loans	Count of active loans where the user is lender.
Active payable loans	Count of active loans where the user is borrower.
Overdue receivables	Outstanding value/count of loans past due where user is lender.
Overdue payables	Outstanding value/count of loans past due where user is borrower.
Due soon	Loans due within configurable next 7/14/30 days.
Pending loan requests	Incoming requests awaiting lender action plus outgoing requests awaiting response.
Repayment confirmations	Repayment claims awaiting the user's action.
Interest expected	Unconfirmed interest portion on active receivable loans.
Completed this month	Count/value of loans completed during the current calendar month, grouped by currency.

- The dashboard shall default to the user's timezone.
- Tapping a metric shall open a filtered list explaining exactly which loans make up the number.
- No cross-currency "grand total" is shown in MVP.
- Metrics shall be computed from authoritative server data, not only from local device state.

15. Notification and Reminder Requirements

Notification	Recipients	Timing
New friend request	Recipient	Immediately
Friend request accepted	Requester	Immediately
New loan request	Lender	Immediately
Loan accepted/rejected	Borrower	Immediately
Bank profile shared	Authorized borrower	Immediately
Due soon	Borrower and lender	Configurable: e.g., 7d, 3d, 1d
Due today	Borrower and lender	Local morning / configured time
Overdue	Borrower and lender	At transition and periodic cadence
Repayment submitted	Lender	Immediately
Repayment confirmed/rejected	Borrower	Immediately

- Push content must avoid exposing sensitive bank information or excessive financial detail on locked screens.
- Every notification type should have an in-app notification record.
- Reminder scheduling must be idempotent to avoid duplicate notifications.
- Timezone changes should trigger rescheduling where materially necessary.

16. Loan History, Search, and Filters

ID	Requirement
HIST-001	Users shall view active, overdue, completed, rejected, cancelled, and disputed loans.
HIST-002	Users shall filter by role, status, friend, due-date range, creation date, and currency.
HIST-003	Users shall search by friend display name, loan reference, and note where privacy policy permits.
HIST-004	Loan details shall present terms, event timeline, repayment history, shared bank profile status, and key timestamps.
HIST-005	Archived history shall remain readable after friendship removal or account deletion, subject to retention policy.

17. Audit Trail and Evidence Requirements

- Create an append-only loan event for request creation, term proposal, acceptance, due-date changes, reminder milestones, repayment submission, repayment confirmation/rejection, dispute events, and administrative interventions.
- Store actor user ID, event type, server timestamp, relevant object IDs, and safe structured metadata.
- Never place raw secrets, full bank account numbers, passwords, tokens, or unnecessary personal content into audit metadata.
- Provide a human-readable timeline derived from the event stream.
- Support export of a single loan summary in a future release without making the event log directly editable.

18. Privacy, Security, and Data Protection

Area	Requirement
Transport	TLS for all client-server and service-to-service network traffic.
Authentication	Short-lived access tokens with secure refresh/session controls or equivalent.
Authorization	Object-level access checks on every loan, repayment, bank profile, share, and friendship API.
Data at rest	Managed database encryption plus field-level encryption for high-sensitivity bank identifiers.
Device storage	Use platform secure storage for tokens; do not store full bank account identifiers in insecure async storage.
Privacy	Collect minimum data needed; define retention and deletion handling for shared financial records.
Abuse controls	Rate limits, blocking, report/dispute mechanisms, and suspicious activity monitoring.
Backups	Automated database backup, restore testing, and documented recovery objectives.
Secrets	Use secret manager/environment injection; no credentials embedded in app bundle or source control.

19. Non-Functional Requirements

ID	Requirement
NFR-001 Availability	Target 99.9% monthly API availability after production stabilization, excluding announced maintenance.
NFR-002 Performance	Typical authenticated read endpoints should target p95 < 500 ms at the API layer under expected load.
NFR-003 Scalability	Application tier should remain stateless so additional instances can be added horizontally.
NFR-004 Reliability	Financial state-changing operations must be transactional and idempotent where retries are possible.
NFR-005 Consistency	Dashboard and loan details must reconcile with confirmed transaction records and loan state.
NFR-006 Maintainability	Backend organized by bounded modules with automated tests, migrations, typed schemas, and code review.
NFR-007 Security	Follow OWASP mobile/API best practices and perform dependency/scanner checks in CI.
NFR-008 Observability	Structured logs, metrics, tracing/correlation IDs, error monitoring, and alerting for failed jobs.
NFR-009 Recovery	Define production RPO/RTO; recommended starting target RPO <= 24h and RTO <= 4h for early MVP, tightened as usage grows.
NFR-010 Compatibility	Support current major iOS/Android versions within the React Native support policy chosen by the team.

20. Error Handling and Edge Cases

Scenario	Required Behavior
Duplicate accept tap	Idempotent backend returns existing accepted result; does not create two loans.
Borrower and lender act simultaneously	Transaction/row locking or optimistic concurrency prevents invalid state transitions.
Loan due while offline	Server changes status/reminders; client reconciles on next sync.
Push delivery fails	In-app notification remains; delivery failure is observable and retrievable where appropriate.
Bank profile revoked	New retrieval denied immediately; loan history notes that payment profile sharing was revoked.
Currency mismatch in repayment	Reject submission; repayment currency is derived from the loan.
Deleted user with completed loans	Shared record remains pseudonymized/retained according to policy rather than corrupting counterpart history.
Friend blocked during active loan	Communication features restricted, but both retain necessary access to the shared debt record and resolution flow.

21. Accessibility and Internationalization

- Use accessible labels, focus order, large-text support, sufficient contrast, and touch targets appropriate for iOS and Android accessibility services.
- All monetary values should be formatted using locale-aware rules while preserving exact currency code.
- Date/time presentation must use user timezone and locale; server storage uses UTC timestamps.
- String resources shall be externalized so localization can be added without screen rewrites.
- Right-to-left layout support should be considered in component design even if not translated at launch.
- Do not rely on color alone for overdue, pending, or completed states.

22. Administrative and Support Requirements

- Admin can suspend accounts, review reports, inspect sanitized event timelines, and view notification delivery diagnostics.
- Admin financial record changes require a dedicated resolution action that writes an immutable administrative event.
- Support tools should hide bank identifiers by default and require elevated permission plus reason to reveal sensitive details.
- User-facing support ticket links should include loan reference IDs but not secret values.
- Feature flags should allow staged rollout of partial repayments, attachments, or new interest methods.

23. Analytics and Observability

- Track registration completion, friend request funnel, loan request funnel, acceptance rate, overdue count, repayment confirmation latency, notification delivery, and feature usage.
- Analytics must use internal identifiers and avoid full names, notes, phone numbers, emails, or bank details in event payloads.
- Technical metrics should include API latency, error rate, queue delay, job failure count, push-provider failure codes, database pool saturation, and cache health.
- Define dashboards and alerts for spikes in failed loan transitions or reminder jobs.

24. Legal, Compliance, and Product Boundaries

- The product must clearly state that it is a record-keeping and reminder tool unless and until a regulated lending/payment model is intentionally introduced.
- Interest, debt enforceability, disclosures, consumer credit, usury, licensing, privacy, and record-retention obligations vary by jurisdiction and require local legal review before launch.
- The app should require users to acknowledge that terms are entered by users and that the platform does not verify that funds were actually transferred.
- If future features include payment processing, credit scoring, debt collection, lending from platform funds, or lending marketplace matching, the compliance model must be re-evaluated before implementation.

Product boundary: The safest MVP is a shared ledger and reminder application. Avoid marketing claims that imply the platform guarantees repayment, validates debt, or acts as a bank/lender.

25. Acceptance Criteria and Release Scope

- A new user can register, verify, sign in, create a profile, and add a friend.
- A borrower can request a loan and a lender can define terms and accept it.
- Both users see identical accepted principal, currency, interest basis, total expected repayment, and due date.
- Dashboard values reconcile with active/overdue loans by currency and role.
- The lender can share a preferred bank profile with the borrower for a specific loan.
- Due reminders are scheduled and recorded for both parties.
- The borrower can submit repayment; the lender can confirm; the completed loan appears in history.
- Unauthorized users cannot retrieve another user's loan or bank details.
- All material financial state changes generate auditable events.
- Core flows are covered by automated API/integration tests and critical React Native UI tests.

26. Future Enhancements and Out-of-Scope Items

Future Feature	Rationale / Note
Partial repayments and installments	Useful for real-world repayment behavior; schema should support from start.
APR/compounding interest	Requires clearer financial disclosures and calculation rules.
Payment integration	Would enable in-app transfers but introduces provider, KYC, compliance, and reconciliation complexity.
Receipts / proof attachments	Useful evidence; requires storage, malware scanning, retention, and moderation considerations.
Loan agreement PDF	Generates a human-readable snapshot of accepted terms and events.
Contact sync / QR friend adding	Faster onboarding with consent/privacy safeguards.
FX conversion dashboard	Needs trusted rates and clear “estimated” labeling; never replaces original-currency ledger.
Trust/reputation score	High-risk product feature; should not be added without fairness, abuse, and legal analysis.
Web dashboard	Optional companion experience for history and exports.
Recurring/installment schedule	Creates scheduled obligations and richer reminder logic.